Critical Safety Warning: Destructive Identity Resets
Before executing any destructive identity resets (such as clearing TPM, deleting computer objects from Microsoft Entra ID/Active Directory, or running dsregcmd /leave), **always verify that BitLocker recovery keys are successfully backed up** to Microsoft Entra ID, Active Directory, or an authorized secure escrow location. Clearing the TPM or purging device certificates without a backed-up recovery key will trigger a permanent BitLocker recovery lockout on reboot.

1. Architecture & Prerequisites Checklist

Identity & Directory

- **Microsoft Entra Connect Sync Scope:** Ensure computer objects requiring co-management are strictly within the Entra Connect synchronization scope. User objects must sync successfully, and the `userCertificate` attribute synchronization must be enabled to support device authentication certificates (`MS-Organization-Access`).

- **Service Connection Point (SCP):** Configure the SCP in the on-premises Active Directory configuration naming context (`CN=Configuration,DC=domain,DC=com`) pointing to the verified Microsoft Entra Tenant ID and Tenant Domain name (`enterprise.transition.manage.microsoft.com`).

- **Network & URL Endpoints:** Ensure firewalls and proxies permit outbound HTTPS traffic (port 443) to Microsoft Entra ID (`*.microsoftonline.com`), Intune enrollment endpoints (`*.manage.microsoft.com`), Windows Push Notification Services (WNS), and Configuration Manager Cloud Management Gateway (CMG) if managing internet-based clients.

Licensing & Scopes

- **Licenses:** Every target user must possess a valid **Microsoft Entra ID P1 or P2** license alongside an **Intune** license (or Microsoft 365 E3/E5 equivalent).

- **MDM User Scope:** Configured inside the Microsoft Intune admin center (`Devices > Enrol devices > Microsoft Intune enrollment > Automatic Enrollment`). Set the **MDM user scope** to **All** or a targeted security group containing the pilot/production device users. *(Note: User scope takes precedence over computer scope in Hybrid scenarios).*

Infrastructure & Policy

- **Configuration Manager:** Run Configuration Manager Current Branch (supported versions aligned with Microsoft lifecycle, typically 2303 or later).

- **Client Settings:** Under the default or custom Client Settings policy, ensure **Register new Windows 10/11 devices with Microsoft Entra ID** is set to **Yes**.

- **Group Policy Object (GPO) for Auto-Enrollment:**

- *Path:* `Computer Configuration > Policies > Administrative Templates > Windows Components > MDM > Enable automatic MDM enrollment using default Microsoft Entra credentials`

- *Configuration:* Enabled, selecting **Default Microsoft Entra credentials** as the credential type.

2. End-to-End Co-Management Enablement Process

1. **Tenant Discovery & Hybrid Join Phase:**

   - Upon domain join or scheduled task execution, the Windows client queries the on-premises Active Directory SCP to retrieve the tenant GUID and authentication endpoints.

   - The client initiates the Hybrid Microsoft Entra join sequence, registering the device object in Microsoft Entra ID and provisioning the `MS-Organization-Access` certificate into the local machine certificate store (`Cert:\LocalMachine\My`).

2. **MDM Auto-Enrollment Trigger Phase:**

   - The Group Policy sets the auto-enrollment registry keys, which registers a scheduled task in Task Scheduler under `Microsoft > Windows > EnterpriseMgmt > Schedule-Initiated-Enrollment`.

   - The task executes `deviceenroller.exe /c /AutoEnroll` in the context of the currently logged-on user (or SYSTEM context if configured via device-based enrollment).

3. **PRT Authentication & Workload Handshake:**

   - `deviceenroller.exe` uses the user's Primary Refresh Token (PRT) or the computer token to authenticate against the Intune enrollment service (`enrollment.manage.microsoft.com`).

   - Intune issues the MDM device certificate, enrolling the device into Intune management.

   - Simultaneously, the Configuration Manager client evaluates the Co-management policy (`CoManagementHandler.log`), detects successful Intune enrollment, establishes communication with the CMG/Management Point, and begins shifting requested workload authorities (e.g., Client Apps, Software Updates, Endpoint Protection) from ConfigMgr to Intune based on the configured slider percentages or target collections.

3. Detection & Diagnostic Steps

Client-Side Verification via `dsregcmd /status`

Run PowerShell as Administrator and execute `dsregcmd /status`. Verify the following output fields:

- `AzureAdJoined : YES` (Confirms Microsoft Entra join status)
- `DomainJoined : YES` (Confirms on-premises AD integration)
- `WamDefaultSet : YES` (Confirms Web Account Manager tokens are functional)
- `MDMUrl : [https://enrollment.manage.microsoft.com/](https://enrollment.manage.microsoft.com/)...` (Must point to the valid Intune enrollment endpoint; if blank, auto-enrollment has not fired).

Essential Client-Side Logs

- **ConfigMgr Co-Management Log:** `C:\Windows\CCM\Logs\CoManagementHandler.log` (Inspect for policy download, enrollment status evaluation, and workload switching codes).
- **ConfigMgr Startup Log:** `C:\Windows\CCM\Logs\ClientIDManagerStartup.log` (Validates SMS GUID and certificate generation).
- **Windows EnterpriseMgmt Logs (Event Viewer):** `Applications and Services Logs > Microsoft > Windows > DeviceManagement-Enterprise-Diagnostics-Provider > Admin` (Look for event ID 75 for successful MDM enrollment or error codes for failure).
- **User Device Registration Logs (Event Viewer):** `Applications and Services Logs > Microsoft > Windows > User Device Registration > Admin` (Verifies PRT acquisition and Hybrid join status).

Console Verification Checks

- **Configuration Manager Console:** Navigate to `Administration > Overview > Cloud Services > Co-management`. Verify that the tenant is connected and target collections show devices reporting as co-managed.
- **Microsoft Intune Admin Center:** Navigate to `Devices > Windows > Windows devices`. Confirm the device record displays **Co-managed** under the Management column.

4. Common Failure Scenarios, Remediation & Verification

Scenario A: Device is Hybrid Joined, but `MDMUrl` is Blank / Intune Enrollment Fails

- **Root Cause:** GPO not applying, user lacks an Intune license, or the MDM user scope is set to "None" or misconfigured security group.
- **Remediation Commands & Steps:**

1. Force group policy update:

   DOS

   ```
   gpupdate /force
   ```

2.

3. Check scheduled task execution status:

   DOS

   ```
   schtasks /run /tn "\Microsoft\Windows\EnterpriseMgmt\Schedule-Initiated-Enrollment"
   ```

4.

5. Verify registry injection for auto-enrollment:

   PowerShell

```
Get-ItemProperty -Path "HKLM:
\SOFTWARE\Policies\Microsoft\Windows\CurrentVersion\MDM"
-Name "AutoEnrollMDM"
```

6.

- **Post-Fix Verification:** Run `dsregcmd /status` and confirm `MDMUrl` is populated with a valid enrollment URL.

Scenario B: Corrupted Identity, Stale Certificates, or PRT Authentication Failures

- **Root Cause:** Orphaned device objects, expired `MS-Organization-Access` certificates, or invalid WAM states resulting in error codes `0x801c0003` (Not Found), `0x80180018` (Enrollment Failed), or `0x80192ee7` (DNS/Network resolution failure).
- **Remediation Commands & Steps:**
  1. Clear corrupt enrollment tokens and force re-evaluation:

     PowerShell

```
# Remove existing scheduled enrollment tasks if hung
Get-ScheduledTask -TaskName "*Enrollment*" | Unregister-
ScheduledTask -Confirm:$false
```

  2.
  3. Clean up stale machine certificates in certmgr (`Cert:\LocalMachine\My`) matching issuer `MS-Organization-Access`.

5. Re-initialize device state:

   ```
   DOS
   ```

   ```
   dsregcmd /leave
   ```

6. *(Reboot the machine to trigger a clean Hybrid Microsoft Entra Join).*

- **Post-Fix Verification:** Run `dsregcmd /status` to confirm `AzureAdJoined` and `DomainJoined` return `YES`.

Scenario C: Co-Management Capability Flags Stuck or Conflicting Authority States

- **Root Cause:** Registry corruption or leftover flags from failed pilots causing desynchronization between ConfigMgr and Intune authorities.
- **Remediation Registry Paths & Keys:** Inspect and adjust keys under:
  - `HKLM\SOFTWARE\Microsoft\CCM\CoManagementFlags`
  - `HKLM\SOFTWARE\Microsoft\Enrollments\<GUID>` (Locate the subkey matching the Intune provider ID and inspect `EnrollmentState`).
- **Remediation Steps:**
  - Stop the SMS Agent Host service:

    ```
    DOS
    ```

    ```
    Stop-Service CcmExec
    ```

    - Clean up corrupted co-management registry configurations under `HKLM\SOFTWARE\Microsoft\CCM\MRM`.
    - Restart the service:

      ```DOS
      Start-Service CcmExec
      ```

- **Post-Fix Verification:** Review `C:\Windows\CCM\Logs\CoManagementHandler.log` for clean initialization and successful policy parsing without exception codes.

Scenario D: ConfigMgr Workload Sliders Not Shifting to Intune

- **Root Cause:** Collection membership mismatch, tenant onboarding token expiration, or client policy evaluation interval delay.
- **Remediation Steps:**
    1. Trigger a machine policy retrieval cycle on the client:

       ```PowerShell
       Invoke-WmiMethod -Namespace root\ccm -Class SMS_Client
       ```

```
-Name TriggerSchedule
"{00000000-0000-0000-0000-000000000021}"
```

2.

3. Verify that the target device is a member of the pilot collection defined in the ConfigMgr Co-Management properties.

- **Post-Fix Verification:** Open `CoManagementHandler.log` and look for entries confirming workload authority shift (e.g., `Transitioning workload 'ClientApps' to Intune`). Validate in the Intune portal that workload management has transitioned successfully.